

Phase I/II Operating Characteristics

Cycle 1 versus Cycle 2 comparison - N = 30, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
One-stage, Cycle_Max = 1	10.2	6.7	3.7	0.6	0.1	78.7
One-stage, Cycle_Max = 2	10.1	8.1	4.3	0.9	0.1	76.5
Two-stage (highest utility), Cycle_Max = 1	10.3	8.5	3.3	0.6	0	77.3
Two-stage (highest utility), Cycle_Max = 2	10	9.8	3.9	0.8	0	75.5
Two-stage (top-2 randomized), Cycle_Max = 1	11.4	7.4	3	0.5	0.1	77.6
Two-stage (top-2 randomized), Cycle_Max = 2	12.1	7.3	2.8	1.1	0	76.7
Mean patients treated						
One-stage, Cycle_Max = 1	11.7	4.1	1.5	0.6	0.1	
One-stage, Cycle_Max = 2	11.7	4.3	1.7	0.7	0.2	
Two-stage (highest utility), Cycle_Max = 1	11.6	4.2	1.7	0.7	0.2	
Two-stage (highest utility), Cycle_Max = 2	11.6	4.4	1.8	0.7	0.2	
Two-stage (top-2 randomized), Cycle_Max = 1	11.4	4.4	1.8	0.7	0.1	
Two-stage (top-2 randomized), Cycle_Max = 2	11.3	4.6	1.9	0.7	0.2	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
One-stage, Cycle_Max = 1	29.4	23.1	20.8	16.9	9.8	0
One-stage, Cycle_Max = 2	28.6	23.4	19.6	17.6	10.8	0
Two-stage (highest utility), Cycle_Max = 1	30.4	23	19.1	16.9	10.6	0
Two-stage (highest utility), Cycle_Max = 2	30.5	23.7	18	15.9	11.9	0
Two-stage (top-2 randomized), Cycle_Max = 1	32.5	25.4	15.8	16.4	9.8	0.1
Two-stage (top-2 randomized), Cycle_Max = 2	33	27.3	17.2	13.2	9.2	0.1
Mean patients treated						
One-stage, Cycle_Max = 1	7.9	6.9	5.7	5.4	4.1	
One-stage, Cycle_Max = 2	8	6.8	5.7	5.3	4.2	
Two-stage (highest utility), Cycle_Max = 1	7.8	6.8	5.7	5.4	4.3	
Two-stage (highest utility), Cycle_Max = 2	7.7	6.8	5.7	5.4	4.4	
Two-stage (top-2 randomized), Cycle_Max = 1	7.7	6.9	5.6	5.6	4.2	
Two-stage (top-2 randomized), Cycle_Max = 2	7.7	6.8	5.6	5.5	4.4	

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Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
One-stage, Cycle_Max = 1	8.6	50.6	23.4	11	2.6	3.8
One-stage, Cycle_Max = 2	10.2	49.6	23.3	10.9	2.6	3.4
Two-stage (highest utility), Cycle_Max = 1	8.8	51.7	24.6	9.5	2.1	3.3
Two-stage (highest utility), Cycle_Max = 2	9.8	48.7	25.2	11.3	2.2	2.8
Two-stage (top-2 randomized), Cycle_Max = 1	7.5	55.2	22.5	10	1.8	3
Two-stage (top-2 randomized), Cycle_Max = 2	7.1	53.2	22.7	11	3.4	2.6
Mean patients treated						
One-stage, Cycle_Max = 1	6.3	11.2	6.7	3.9	1.3	
One-stage, Cycle_Max = 2	6.3	11	6.7	4.1	1.4	
Two-stage (highest utility), Cycle_Max = 1	6	10.6	7.2	4.2	1.5	
Two-stage (highest utility), Cycle_Max = 2	6	10.2	7.2	4.5	1.7	
Two-stage (top-2 randomized), Cycle_Max = 1	7.1	9.9	7.1	4.2	1.4	
Two-stage (top-2 randomized), Cycle_Max = 2	7	9.5	7.2	4.3	1.6	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
One-stage, Cycle_Max = 1	1.4	2.1	5.7	59.2	31.6	0
One-stage, Cycle_Max = 2	1.4	1.5	5.2	58.8	33.1	0
Two-stage (highest utility), Cycle_Max = 1	1.7	1.1	5.7	56.6	34.9	0
Two-stage (highest utility), Cycle_Max = 2	1.5	1.6	6	55.7	35.2	0
Two-stage (top-2 randomized), Cycle_Max = 1	0.3	0.7	3.5	54.4	41.1	0
Two-stage (top-2 randomized), Cycle_Max = 2	0.8	1	4.4	56.9	36.9	0
Mean patients treated						
One-stage, Cycle_Max = 1	3.5	3.5	4.4	12.4	6.1	
One-stage, Cycle_Max = 2	3.5	3.5	4.4	12.3	6.3	
Two-stage (highest utility), Cycle_Max = 1	3.5	3.5	4.2	10.8	7.9	
Two-stage (highest utility), Cycle_Max = 2	3.5	3.5	4.2	10.7	8.2	
Two-stage (top-2 randomized), Cycle_Max = 1	4	4	4.8	9.2	8	
Two-stage (top-2 randomized), Cycle_Max = 2	3.9	4	4.9	9.2	8	

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Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
One-stage, Cycle_Max = 1	14.1	25.3	42	10.6	0.8	7.2
One-stage, Cycle_Max = 2	14.1	24	44	11	0.7	6.2
Two-stage (highest utility), Cycle_Max = 1	14	25.1	43.9	9.4	1	6.6
Two-stage (highest utility), Cycle_Max = 2	15.1	22.4	44.8	10.6	1.3	5.8
Two-stage (top-2 randomized), Cycle_Max = 1	12.2	26.8	42.5	11.3	1.1	6.1
Two-stage (top-2 randomized), Cycle_Max = 2	11.6	24.2	44.8	12.7	1.1	5.6
Mean patients treated						
One-stage, Cycle_Max = 1	8.1	7.7	9.2	3.4	0.6	
One-stage, Cycle_Max = 2	7.9	7.5	9.4	3.6	0.6	
Two-stage (highest utility), Cycle_Max = 1	7.5	7.7	9.3	3.7	0.9	
Two-stage (highest utility), Cycle_Max = 2	7.4	7.4	9.3	4	1.1	
Two-stage (top-2 randomized), Cycle_Max = 1	8.5	8.3	7.9	3.6	0.9	
Two-stage (top-2 randomized), Cycle_Max = 2	8.3	8.1	8	3.9	1.1	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
Utility 3	41	40.8	42.4	43	58	
OBD selection (%)						
One-stage, Cycle_Max = 1	9.4	10.7	10.7	8.3	59.5	1.4
One-stage, Cycle_Max = 2	8.9	10.9	9.1	9.5	60.4	1.2
Two-stage (highest utility), Cycle_Max = 1	9.7	11.3	11.5	8.7	57.8	1
Two-stage (highest utility), Cycle_Max = 2	10.1	11	8.9	9	60.1	0.9
Two-stage (top-2 randomized), Cycle_Max = 1	10.2	9.8	9.8	9.8	59.2	1.2
Two-stage (top-2 randomized), Cycle_Max = 2	9.2	10.8	9.3	8.9	60.8	1
Mean patients treated						
One-stage, Cycle_Max = 1	5.6	5.2	4.6	4.8	9.6	
One-stage, Cycle_Max = 2	5.6	5	4.5	4.8	9.9	
Two-stage (highest utility), Cycle_Max = 1	5.5	5.1	4.5	4.9	9.8	
Two-stage (highest utility), Cycle_Max = 2	5.4	5	4.5	4.9	10	
Two-stage (top-2 randomized), Cycle_Max = 1	6.6	6.1	5.1	5.1	7.1	
Two-stage (top-2 randomized), Cycle_Max = 2	6.5	6	5	5	7.4	